

# LIN7209 – Syntax

## Extended projections

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Adèle Hénot-Mortier (based on David's original materials)

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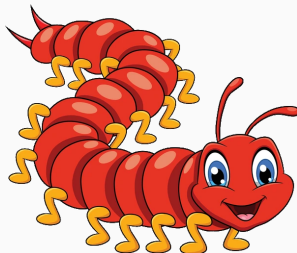
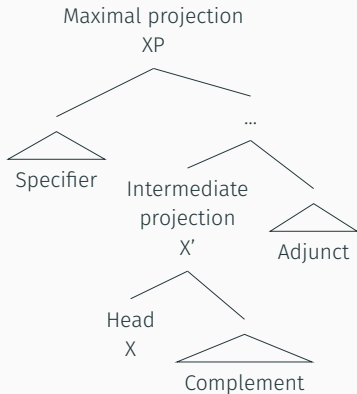
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# Plan for today

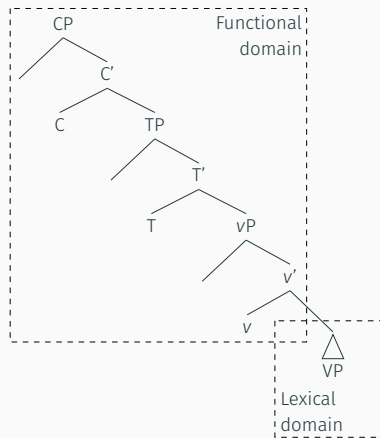
- Review of phrase structure, types of heads,  $\theta$ -roles.
- Extended projections: description and advantages.

# Heads and structural relations

- Constituents tend to inherit the distribution and properties of one of their elements—the **head**.
- The other elements of the constituent can then be defined by their **structural relation** to the head.



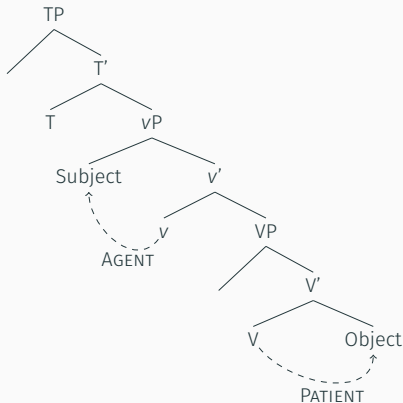
# Functional vs. lexical



- **Lexical** heads are open-class, semantically varied elements. Examples: verbs, nouns, adjectives...
- **Functional** heads are closed-class, phonologically weak, semantically fixed, “functional”/“relational” elements. Examples: complementizers, tense, light verbs (DO, CAUSE...).
- At the clausal level, functional heads tend to be **higher** than lexical ones.

# The verbal domain

- The complement (object) of the lexical verb head (V) is its only true argument. This **object** often gets a **PATIENT**  $\Theta$ -role.
- *v* is a functional head which hosts a “light” verb (CAUSE, Do), covert in English. *v* assigns a  $\Theta$ -role to its specifier (the **subject**), often **AGENT**.



## Morphological evidence for v

- Malagasy (Austronesian), Hiaki (Uto-Aztecan), Japanese (Japonic) have an affix for v meaning “active voice”/CAUSE.

(1) Malagasy:

M-**an**-sasa ny lamba amin ny savony Rasoa  
PST-**v**-wash the clothes with the soap Rasoa

‘Rasoa causes the clothes to be clean (=washes) with the soap.’

(2) Hiaki:

Huan u’usit-ta ee-**tua**-k.  
Juan child-ACC feel-**v**-PST

‘Juan teased the child.’

(3) Japanese:

Keiko-wa pizza-o ag-**e**-ta  
Keiko-TOP pizza-ACC rise-**v**-PST

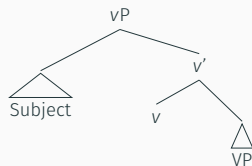
‘Keiko raised the pizza.’

## Semantic evidence for $v$

- If a given constituent is a true argument of a head, then the head may be sensitive to its semantic characteristics.
- The interpretation of  $V$  is often sensitive to the semantic characteristic of its object, but never to those of its subject!

- (4) a. **throw** a baseball. throw  $\simeq$  cast  
b. **throw** support behind a candidate. throw  $\simeq$  provide  
c. **throw** a party. throw  $\simeq$  organize

- This implies subjects are not true arguments of the verb, and motivates a structure whereby subjects are not specifiers of  $V$ , but instead specifiers of a higher functional head, namely  $v$ .



# Testing for AGENCY

- Purpose clauses.

(5)     Jo smiled in order to reassure Ed.  
         AGENT

(6)   ??   Jo     heard the conversation to gain new insights.  
         EXPERIENCER

(7)   ??     Jo     fell in order to distract the police.  
         PATIENT/THEME

- Adverbs like *deliberately*, *intentionally*

(8)   Jo {smiled / ??heard the conversation / ??fell } deliberately.

- Passive alternation: AGENT can be in the *by*-phrase



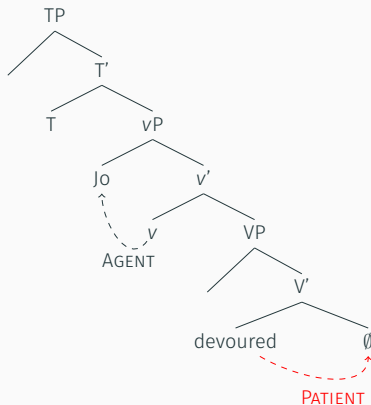
## More thematic roles (THEME is a little bit of a catch-all)

- (9) a. Jo fell. Unaccusative  
PATIENT/THEME
- b. Jo jumped. Unergative  
AGENT
- (10) a. Jo melted the ice. Causative  
AGENT THEME
- b. The ice melted. Inchoative  
THEME
- (11) a. Jo fears snakes. Psych verb  
EXPERIENCER STIMULUS/THEME
- b. Snakes frighten Jo. Psych verb  
STIMULUS/THEME EXPERIENCER
- (12) a. The book is blue. Copular/Predicative construction  
THEME
- b. The ball rolled away. Change-of-location  
THEME
- c. Jo gave Ed the book. Ditransitive  
Agent GOAL THEME

- Which role(s) get(s) assigned depends on the semantics of the verb, defining its “ $\Theta$ -grid”.

# Missing $\Theta$ -roles

(13) \*Jo devoured.

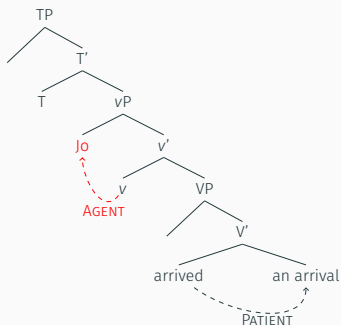


- If a strongly transitive verb does not have an object, a  $\Theta$ -role is unassigned, and the  $\Theta$ -grid of the verb is not “exhausted”. Ungrammaticality arises.

# Too many arguments

(14) Jo arrived \*(an arrival).

Unaccusative



- If the *v* in unaccusative structures assigned an AGENT  $\Theta$ -role, then we could not explain why such structures are happy with one single PATIENT argument (see previous slide).
- So, *v* must be defective (empty  $\Theta$ -grid), and then *Jo* gets no  $\Theta$ -role... causing ungrammaticality.

# Constraints on $\Theta$ -roles

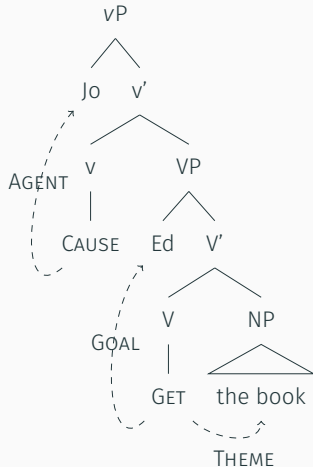
- The  **$\Theta$ -criterion**:<sup>1</sup> each  $\Theta$ -role gets assigned to one and only one argument, and each argument receives one and only one  $\Theta$ -role.
- The **Uniformity of Theta-Assignment Hypothesis** (UTAH):<sup>2</sup>  
Identical **thematic** relationships between items are represented by identical **structural** relationships between those items at deep structure (i.e. before movement/INTERNAL MERGE).
  - AGENT = causer = Spec-vP
  - PATIENT = undergoer = Comp-V
- $\Theta$ -roles are assigned **locally**, typically in a Comp-Head or Spec-Head configuration.

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<sup>1</sup>Chomsky, 1981.

<sup>2</sup>Baker, 1988, 1997.

## Ditransitives (Adger, 2003)



(15) Jo gave Ed the book.

- Give analyzed as **CAUSE (light verb, v)** to **GET (lexical verb, V)**.
- v gets has one argument (**AGENT**), and V two arguments (**GOAL** and **THEME**).
- More complex analyses may assume an extra functional projection between **vP** and **VP**, assigning its role to the **GOAL** argument.
- Backed by languages like Kinyarwanda (Bantu), in which this extra projection gets an overt “applicative” head.

## Extended projections

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# Comparing verbs and nouns

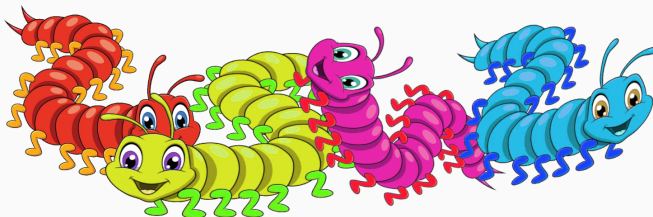
- Verbs and nouns assign very similar  $\Theta$ -roles.

- (16) a. Jo 's **interview** of the candidate  
AGENT PATIENT/THEME
- b. Jo 's **love** of music  
EXPERIENCER STIMULUS/THEME
- c. Jo's **mother**.  
Kin
- d. A **cup** of tea, the **top** of the desk, the **King** of France,  
**paintings** of Jo...

- Both verbs and nouns generally come with a bunch of other projections above them: CP, TP, vP for verbs, PP, DP for nouns.
- A lot of symmetries between verbs and nouns! How to cash this out?

# Assigning structure to structure

- Idea: we don't have isolated projections (the “centipedes”), we have a bunch of them having a big party together.
- Groups of related projections are called **extended projections**.<sup>3</sup>
- Main advantage: explain why relations between projection that are expected to be **local** (between adjacent projections), sometimes appear **non-local** (i.e. seemingly “skip” projections).



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<sup>3</sup>Grimshaw, 1991.

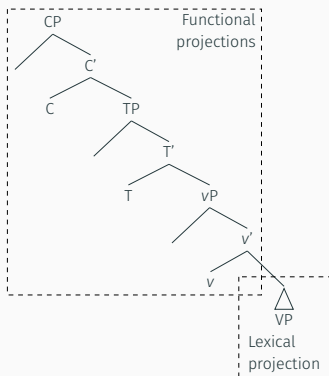


# Extended projections



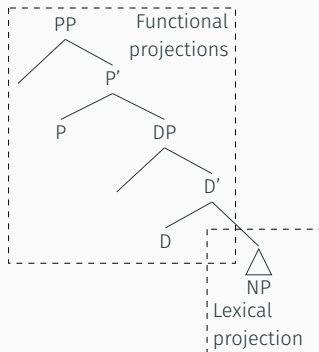
- Extended projections will typically involve a **lexical projection downstairs**, and one or more **functional projections upstairs**.
- All projections (functional/lexical) belong to the same overall “**category**”/**domain**: typically verbal or nominal.
- Within an extended projection, information (features, headedness) can circulate, s.t. the higher functional projection can display the features of a lower lexical head belonging to the same extended projection.

# Extended projection in the V-domain



- V is a lexical head.
- vP, TP, CP are functional projections stacked over it.
- $\{CP, TP, vP, VP\}$  forms an **extended projection of the verbal category**.
- In this extended projection, the lexical projection VP is downstairs, the functional ones upstairs.

# Extended projection in the N-domain



- N is a lexical head.
- DP, PP are functional projections that can be stacked over it.
- $\{PP, DP, NP\}$  forms an **extended projection of the nominal category**.
- In this extended projection, the lexical projection NP is downstairs, the functional ones upstairs.
- Note how the V- and the N-domains have the same core organization!

## Arguments for an extended projection in the nominal domain

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## V-heads can access N-features across the D projection

- Selection: verbs can select certain features on the noun, **even in the presence of an intervening determiner** forming its own projection!

(17) They merged [<sub>DP</sub> the [<sub>NP</sub> files / \*file / dogs]].

- Agreement: sensitive to the features of the head, again despite intervening determiners!

(18) [<sub>DP</sub> The [<sub>NP<sub>[PL]</sub></sub> dogs]] {is/\*are}...

(19) [<sub>DP</sub> [<sub>DP</sub> The boy] [<sub>D</sub> 's] [<sub>NP<sub>[SG]</sub></sub> book]] {is/\*are}...

- Problem solved if we assume that DP and NP are part of the same projection, s.t. DP can “display” N’s features.
- **What seemed to be agreement/selection “at a distance” is in fact local**, as soon as extended projections are considered.

# Pied-piping in *wh*-questions



- *Wh*-movement: we expect the *wh*-phrase (a DP) to be fronted.
- However, it's also possible to front the entire PP, whose complement is the *wh*-phrase! This is called **pied-piping**.
- It's also active in English relative clauses.

- (20)
- a. They found a note [<sub>PP</sub> under [<sub>DP</sub> **that stone**]].
  - b. [<sub>DP</sub><sub>[+wh]</sub> **Which stone**] did they find a note [<sub>PP</sub> under \_\_\_]?
  - c. [<sub>PP</sub> Under [<sub>DP</sub><sub>[+wh]</sub> **which stone**]] did they find a note \_\_\_?

# Pied-piping and Extended Projections

(20c)  $\frac{[PP_{[+wh]} \text{ Under } [DP_{[+wh]} \text{ which stone}]]}{\text{Extended Projection}}$  did they find a note \_\_?

- Problem solved if we assume that DP and PP are part of the same projection, s.t. PP can “display” *wh*-properties and become the target of *wh*-movement.
- What seemed to be the *wh*-movement of something “too big” (the PP) is in fact equivalent to the movement of its complement DP, as soon as extended projections are considered.
- **Pied-piping is equivalent to DP-level *wh*-movement given Extended Projections.**

## V-heads can access the +wh feature across the P projection

- Extra evidence for the fact that PP belongs to N's extended projection comes from the selection of interrogative complements by verbs like *wonder*.
- Normally, this selection is super-local: the highest projection must bear the [+wh] feature.
- Yet, Ps **can intervene between the selecting verb and the wh-projection**.

- (21) a. I wonder [<sub>CP</sub> [<sub>DP<sub>[+wh]</sub></sub> whose book] they read].  
b. \* I wonder [<sub>CP</sub> [<sub>DP</sub> a book] about [<sub>DP<sub>[+wh]</sub></sub> who] they read].  
c. I wonder [<sub>CP</sub> [<sub>PP<sub>[+wh]</sub></sub> by [<sub>DP<sub>[+wh]</sub></sub> who]] the book was written].

- Again problem solved if PP and DP are part of the same extended projection, and PP can “display” its complement DP's wh-properties.



## Arguments for an extended projection in the verbal domain

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# Mood selection

- Certain verbs selecting for the same complementizer, **force distinct moods** on the embedded verb...
- This is surprising under a purely local account: we'd not expect the embedding V to mess up with the embedded T (hosting MOOD) across C, plus **we'd expect the same Cs to select for the same Ts!**

- (22) a. Jo requested [<sub>CP</sub> **that** [<sub>TP</sub> Ed {leave.SBJV/?left.IND} at 6]].  
b. Jo thought [<sub>CP</sub> **that** [<sub>TP</sub> Ed {\*leave.SBJV/left.IND} at 6]].

- Problem solved if we assume that CP and TP are part of the same projection, s.t. CP can “display” T's features (granted they include mood).
- What seemed to be selection “at a distance” is in fact local, as soon as extended projections are considered.

# References i



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